General Engineering Assessment - Alternative Power for C-Star

Role: General Engineer
Deliverable: A note of no more than 3 pages, plus any supporting work (e.g spreadsheet /model/script) you built.

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1. The situation

A C-Star is a small autonomous sailing robot designed for long-duration ocean missions.

A customer is interested in using the C-Star at high latitudes, where solar power may be limited. They have asked whether adding either a small wind turbine or a wave-energy harvester could meaningfully improve the C-Star's endurance.

You have 3 hours to investigate the problem and recommend what Oshen should do.

You can use internet research, calculations, AI tools and any other tools you would normally use at work. Where information is missing, make reasonable assumptions or identify what you would need to find out.

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2. What you're given

Platform

For this exercise, assume approximately the following:

- 1 m long, 40 kg
- Rigid wingsail, approximately 0.6 m² surface area
- 50 W peak solar generation
- 1300 Wh $LiFePO_4$ battery
- Approximately 1 W average electrical consumption
- Onboard computing, communications, navigation, control and sensors

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3. Results for you to provide

Assess whether either concept looks technically worthwhile.

- whether you think it could work and why;
- the main engineering challenges or blockers;
- how you would integrate it with C-Star;
- how you would test the idea before investing significantly in it; and
- what you would recommend Oshen does next.

Consider how your approach would differ if you had six months to go from concept design to prototype manufacturing vs two months.

We are not expecting a finished design by any means. We are interested in how you research, reason about and de-risk an unfamiliar engineering problem.

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4. Constraints on how you work

- Use whatever tools you normally use at work, including AI, we don't care
- Three pages maximum. Send any supporting work (spreadsheet / script) as a separate file - we will open it.
- Where you don't have information, make an assumption, label it as one, and say what it would take to firm it up.